

LDL06F02B

2008 年 9 月 27 日

第 1 章

論理譜

```

{ ===== }
{   メモリ   }
{ ===== }
logicname LDL06F02B

{ ===== }
{   4 ビットメモリ   }
{ ===== }

procedure MEMORY
input  RESET;
input  CS;
input  WP;
input  DATAIN[4];
output DATAOUT[4];

bitr memory[4];

    DATAOUT=memory;

    if (RESET)
        memory=0;
    else
        if (WP)
            if (CS)
                memory=DATAIN;
            else
                memory=memory;
            endif
        else
            memory=memory;
        endif
    endif
endp

```

```

{ ===== }
{ 実効譜 }
{ ===== }
entity main
{ ----- }
{ 入力 }
{ ----- }
input RESET;
input ADDA[3];
input ADDB[3];
input ADDC[3];
input DATAIN[4];
input DATAIN[4];
input WP;
input WCP;

{ ----- }
{ 出力 }
{ ----- }
output DATAOUT[4];
output DATAOUT[4];
output BLANK[3];

{ ----- }
{ 内部信号 }
{ ----- }
bitn memory0[4];
bitn memory1[4];
bitn memory2[4];
bitn memory3[4];
bitn memory4[4];
bitn memory5[4];
bitn memory6[4];
bitn memory7[4];
bitr wp[2];
bitr mode[2];
bitn we;
bitn mcs[8];
bitn writedata0p[4];
bitn writedata1p[4];
bitn writedata2p[4];
bitn writedata3p[4];
bitn writedata4p[4];
bitn writedata5p[4];
bitn writedata6p[4];
bitn writedata7p[4];
bitr regADDC[3];
bitn blank[3];
bitn memoryzeq[8];

{ ----- }
{ 検査端子 }
{ ----- }

output T0P[2]; T0P=mode;
output T1P[2]; T1P=wp;
output T2P; T2P=we;
output T3P[8]; T3P=mcs;
output T4P[4]; T4P=memory0;
output T5P[4]; T5P=memory1;
output T6P[4]; T6P=memory2;
output T7P[4]; T7P=memory3;
output T8P[4]; T8P=memory4;

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output T9P[4]; T9P=memory5;
output T10P[4]; T10P=memory6;
output T11P[4]; T11P=memory7;

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{ ----- }
{   読み出し A   }
{ ----- }

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switch(ADDA)
case 0: DATAAOUT=memory0;
case 1: DATAAOUT=memory1;
case 2: DATAAOUT=memory2;
case 3: DATAAOUT=memory3;
case 4: DATAAOUT=memory4;
case 5: DATAAOUT=memory5;
case 6: DATAAOUT=memory6;
case 7: DATAAOUT=memory7;
endswitch

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{ ----- }
{   読み出し B   }
{ ----- }

```

```

switch(ADDB)
case 0: DATABOUT=memory0;
case 1: DATABOUT=memory1;
case 2: DATABOUT=memory2;
case 3: DATABOUT=memory3;
case 4: DATABOUT=memory4;
case 5: DATABOUT=memory5;
case 6: DATABOUT=memory6;
case 7: DATABOUT=memory7;
endswitch

```

```

{ ----- }
{   メモリ本体   }
{ ----- }

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memory0=MEMORY(RESET,mcs.0,we,writedata0p);
memory1=MEMORY(RESET,mcs.1,we,writedata1p);
memory2=MEMORY(RESET,mcs.2,we,writedata2p);
memory3=MEMORY(RESET,mcs.3,we,writedata3p);
memory4=MEMORY(RESET,mcs.4,we,writedata4p);
memory5=MEMORY(RESET,mcs.5,we,writedata5p);
memory6=MEMORY(RESET,mcs.6,we,writedata6p);
memory7=MEMORY(RESET,mcs.7,we,writedata7p);

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```

memory0.4=1;
memory1.4=1;
memory2.4=1;
memory3.4=1;
memory4.4=1;
memory5.4=1;
memory6.4=1;
memory7.4=1;

```

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{ ----- }
{   書き込み行程   }
{ ----- }

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if (RESET)
wp=0;
else
switch(wp)
case 0:
switch(WP,WCP)
case 0,0: wp=0;
case 1,0: wp=1; { せり上がり書き込み }
case 0,1: wp=2; { 番地指定書き込み }

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        case 1,1: wp=2; { " }
    endswitch
    case 1: wp=3; { セリ上がり書き込み行程選択 }
    case 2: wp=3; { 番地指定書き込み行程選択 }
    case 3: wp=0; { 書き込み実行 }
    endswitch
endif

{ ----- }
{ 書き込み指標 }
{ ----- }
we=wp==3;

{ ----- }
{ 書き込み種別 }
{ ----- }

if (RESET)
    mode=0;
else
    switch(wp)
        case 0: mode=0;
        case 1: mode=1;
        case 2: mode=2;
        default: mode=mode;
    endswitch
endif

{ ----- }
{ 番地指定書き込み番地記憶 }
{ ----- }

if (RESET)
    regADDC=0;
else
    switch(wp)
        case 0: regADDC=ADDC;
        default: regADDC=regADDC;
    endswitch
endif

{ ----- }
{ メモリ選択 }
{ ----- }

switch(mode)
    case 1:
        mcs=0xff;
    case 2:
        switch(regADDC)
            case 0: mcs.0=1;
            case 1: mcs.1=1;
            case 2: mcs.2=1;
            case 3: mcs.3=1;
            case 4: mcs.4=1;
            case 5: mcs.5=1;
            case 6: mcs.6=1;
            case 7: mcs.7=1;
        endswitch
    endswitch

{ ----- }
{ 書き込みデータ }
{ ----- }

switch(mode)
    case 1:
        writedata0p=DATAIN;
        writedata1p=memory0;

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        writedata2p=memory1;
        writedata3p=memory2;
        writedata4p=memory3;
        writedata5p=memory4;
        writedata6p=memory5;
        writedata7p=memory6;
    case 2:
        writedata0p=DATAACIN;
        writedata1p=DATAACIN;
        writedata2p=DATAACIN;
        writedata3p=DATAACIN;
        writedata4p=DATAACIN;
        writedata5p=DATAACIN;
        writedata6p=DATAACIN;
        writedata7p=DATAACIN;
    endswitch

```

```

{ ----- }
{ 0 指標 }
{ ----- }

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```
memoryzeq.0=0;
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switch(memory1)
    case 0: memoryzeq.1=1;
endswitch

```

```

switch(memory2)
    case 0: memoryzeq.2=1;
endswitch

```

```

switch(memory3)
    case 0: memoryzeq.3=1;
endswitch

```

```

switch(memory4)
    case 0: memoryzeq.4=1;
endswitch

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```

switch(memory5)
    case 0: memoryzeq.5=1;
endswitch

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```

switch(memory6)
    case 0: memoryzeq.6=1;
endswitch

```

```

switch(memory7)
    case 0: memoryzeq.7=1;
endswitch

```

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{ ----- }
{ 0 消灯 }
{ ----- }

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switch (memoryzeq.1:7)
    case 0b1111111: blank=1;
    default:
        switch (memoryzeq.2:7)
            case 0b111111: blank=2;
            default:
                switch(memoryzeq.3:7)
                    case 0b11111: blank=3;
                    default:
                        switch(memoryzeq.4:7)
                            case 0b1111: blank=4;
                            default:

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switch(memoryzeq.5:7)
  case 0b111: blank=5;
  default:
    switch(memoryzeq.6:7)
      case 0b11: blank=6;
      default:
        switch(memoryzeq.7)
          case 1: blank=7;
        endswitch
      endswitch
    endswitch
  endswitch
endswitch
endswitch
BLANK=blank;
ende
```



```

{ ===== }
{ 機能実行譜 }
{ ===== }
entity sim
{ ----- }
{ 検査端子 }
{ ----- }
output RESET;
output ADDA[3];
output ADDB[3];
output ADDC[3];
output DATAIN[4];
output DATACIN[4];
output WP;
output WCP;
output DATAAOUT[4];
output DATABOUT[4];
output BLANK[3];
input simres;

{ ----- }
{ 内部信号 }
{ ----- }
bitr tc[8];

{ ----- }
{ 検証論理引き出し端子 }
{ ----- }
output TBOP[8]; TBOP=tc;

{ ----- }
{ 実効譜導入 }
{ ----- }
part main(RESET,ADDA,ADDB,ADDC,DATAIN,DATACIN,WP,WCP
,DATAAOUT,DATABOUT,BLANK)

{ ----- }
{ 検証計数 }
{ ----- }
simres=0;
if (!simres) tc=tc+1; endif

{ ----- }
{ 初期化 }
{ ----- }
if (tc<5) RESET=1; endif

{ ----- }
{ セリ上がり書き込み }
{ ----- }
switch(tc)
case 10: DATAIN=10; WP=1;
case 11: DATAIN=10; WP=0;
case 12: DATAIN=10; WP=0;

case 13: DATAIN=11; WP=1;
case 14: DATAIN=11; WP=0;
case 15: DATAIN=11; WP=0;

case 16: DATAIN=12; WP=1;
case 17: DATAIN=12; WP=0;
case 18: DATAIN=12; WP=0;

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case 19: DATAIN=14; WP=1;
case 20: DATAIN=14; WP=0;
case 21: DATAIN=14; WP=0;

case 22: DATAIN=15; WP=1;
case 23: DATAIN=15; WP=0;
case 24: DATAIN=15; WP=0;

case 25: DATAIN=9; WP=1;
case 26: DATAIN=9; WP=0;
case 27: DATAIN=9; WP=0;

case 28: DATAIN=8; WP=1;
case 29: DATAIN=8; WP=0;
case 30: DATAIN=8; WP=0;

case 31: DATAIN=7; WP=1;
case 32: DATAIN=7; WP=0;
case 33: DATAIN=7; WP=0;

case 35: ADDB=1; ADDA=7;
case 36: ADDB=2; ADDA=6;
case 37: ADDB=3; ADDA=5;
case 38: ADDB=4; ADDA=4;
case 39: ADDB=5; ADDA=3;
case 40: ADDB=6; ADDA=2;
case 41: ADDB=7; ADDA=1;
endswitch

{ ----- }
{ 番地指定書き込み no.1 }
{ ----- }

switch(tc)
case 50: DATACIN=1; ADDC=0; WCP=1;
case 51: DATACIN=1; ADDC=0; WCP=0;
case 52: DATACIN=1; ADDC=0; WCP=0;

case 53: DATACIN=6; ADDC=5; WCP=1;
case 54: DATACIN=6; ADDC=5; WCP=0;
case 55: DATACIN=6; ADDC=5; WCP=0;

case 56: DATACIN=3; ADDC=2; WCP=1;
case 57: DATACIN=3; ADDC=2; WCP=0;
case 58: DATACIN=3; ADDC=2; WCP=0;

case 59: DATACIN=4; ADDC=3; WCP=1;
case 60: DATACIN=4; ADDC=3; WCP=0;
case 61: DATACIN=4; ADDC=3; WCP=0;

case 62: DATACIN=5; ADDC=4; WCP=1;
case 63: DATACIN=5; ADDC=4; WCP=0;
case 64: DATACIN=5; ADDC=4; WCP=0;

case 65: DATACIN=2; ADDC=1; WCP=1;
case 66: DATACIN=2; ADDC=1; WCP=0;
case 67: DATACIN=2; ADDC=1; WCP=0;

case 68: DATACIN=7; ADDC=6; WCP=1;
case 69: DATACIN=7; ADDC=6; WCP=0;
case 70: DATACIN=7; ADDC=6; WCP=0;

case 71: DATACIN=8; ADDC=7; WCP=1;
case 72: DATACIN=8; ADDC=7; WCP=0;
case 73: DATACIN=8; ADDC=7; WCP=0;

case 75: ADDB=1; ADDA=7;

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        case 76: ADDB=2; ADDA=6;
        case 77: ADDB=3; ADDA=5;
        case 78: ADDB=4; ADDA=4;
        case 79: ADDB=5; ADDA=3;
        case 80: ADDB=6; ADDA=2;
        case 81: ADDB=7; ADDA=1;
    endswitch

{ ----- }
{ 番地指定書き込み no.2 }
{ ----- }

    switch(tc)
        case 100: DATACIN=1; ADDC=0; WCP=1;
        case 101: DATACIN=1; ADDC=0; WCP=0;
        case 102: DATACIN=1; ADDC=0; WCP=0;

        case 103: DATACIN=2; ADDC=5; WCP=1;
        case 104: DATACIN=2; ADDC=5; WCP=0;
        case 105: DATACIN=2; ADDC=5; WCP=0;

        case 106: DATACIN=1; ADDC=2; WCP=1;
        case 107: DATACIN=1; ADDC=2; WCP=0;
        case 108: DATACIN=1; ADDC=2; WCP=0;

        case 109: DATACIN=2; ADDC=3; WCP=1;
        case 110: DATACIN=2; ADDC=3; WCP=0;
        case 111: DATACIN=2; ADDC=3; WCP=0;

        case 112: DATACIN=1; ADDC=4; WCP=1;
        case 113: DATACIN=1; ADDC=4; WCP=0;
        case 114: DATACIN=1; ADDC=4; WCP=0;

        case 115: DATACIN=2; ADDC=1; WCP=1;
        case 116: DATACIN=2; ADDC=1; WCP=0;
        case 117: DATACIN=2; ADDC=1; WCP=0;

        case 118: DATACIN=1; ADDC=6; WCP=1;
        case 119: DATACIN=1; ADDC=6; WCP=0;
        case 120: DATACIN=1; ADDC=6; WCP=0;

        case 121: DATACIN=2; ADDC=7; WCP=1;
        case 122: DATACIN=2; ADDC=7; WCP=0;
        case 123: DATACIN=2; ADDC=7; WCP=0;

        case 130: ADDB=1; ADDA=7;
        case 131: ADDB=2; ADDA=6;
        case 132: ADDB=3; ADDA=5;
        case 133: ADDB=4; ADDA=4;
        case 134: ADDB=5; ADDA=3;
        case 135: ADDB=6; ADDA=2;
        case 136: ADDB=7; ADDA=1;
    endswitch

ende

```

```

{ ===== }
{ 機能実行譜 }
{ ===== }
entity sim
{ ----- }
{ 検査端子 }
{ ----- }
output RESET;
output CS;
output WE;
output DATAIN[4];
output DATAOUT[4];

{ ----- }
{ 内部信号 }
{ ----- }
bitn node_DATAIN[4];
bitn node_RESET;
bitn node_CS;
bitn node_WE;

bitr tc[8];

{ ----- }
{ 検査端子代入 }
{ ----- }
RESET=node_RESET;
CS=node_CS;
WE=node_WE;
DATAIN=node_DATAIN;

{ ----- }
{ 手続き引用 }
{ ----- }
DATAOUT=MEMORY(node_RESET,node_CS,node_WE,node_DATAIN);

{ ----- }
{ 検証計数 }
{ ----- }
tc=tc+1;

{ ----- }
{ 初期化 }
{ ----- }
if (tc<5) node_RESET=1; endif

{ ----- }
{ メモリ操作 }
{ ----- }
switch(tc)
case 10: node_CS=1; node_WE=1; node_DATAIN=5;
case 13: node_CS=1; node_WE=1; node_DATAIN=6;
case 16: node_CS=1; node_WE=1; node_DATAIN=7;
case 19: node_CS=1; node_WE=1; node_DATAIN=8;
endswitch
ende
endlogic

```